R-UNIVERSALITY OF THE SPECIAL LINEAR LIE ALGEBRA OVER A CENTRAL DIVISION ALGEBRA IS EXACTLY THE SPLIT-OR-QUATERNION CONDITION

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Allowed standard prerequisites (editorial).

- The Tits algebras of the natural representation of SL_n(D) are the tensor powers of D. Used inside the counted proof at source lines 550-552 and again at source lines 559-561, attributed by the author to page 378 of the cited book KnusEtAl-BookOfInvols. Identifies the Brauer classes whose behaviour under base change decides R-universality.
- The division-algebra description of R-universality through the Tits algebras. The author's equivalence labelled RunivIff, used inside the counted proof in both directions at source lines 554-556 and 562-563. Turns the Brauer-class computation into the two implications of the statement.

Source reference. Original lemma 2.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 416–416. Navigation pointer only: the counted proof invokes this equivalence in both directions; its content is declared as a named standard prerequisite and it has its own card in this lane.

Source reference. Original definition 4: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 432–432. Navigation pointer only: the labelled item of that equivalence used in the forward direction.

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Source reference. Original definition 3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 428–428. Navigation pointer only: the labelled item of that equivalence used in the converse.

Proposition 0.1. *Let $\mathfrak{g} = \mathfrak{sl}_n(D)$, where D is a central division algebra over $\mathbb{Q}$. Then $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if D is either $\mathbb{Q}$ or a quaternion algebra that does not split over $\mathbb{R}$.*

Proof. Let d be the degree of D over $\mathbb{Q}$, and let λ be the highest weight of the standard representation of $\mathfrak{sl}_{dn}(\mathbb{C})$, so $\{\lambda^i\}_{i=1}^{dn}$ is a set of representatives of Z^*. Then $D_{\mathfrak{g},\mathbb{Q}}(\lambda^i)$ and $D_{\mathfrak{g},\mathbb{R}}(\lambda^i)$ are Brauer equivalent to the i-fold tensor products $D^{\otimes i}$ and $D_{\mathfrak{g},\mathbb{R}}(\lambda)^{\otimes i}$, respectively [4, p. 378].

If $\mathfrak{g}$ is $\mathbb{R}$-universal, then, by taking $i = 1$ and noting that $D_{\mathfrak{g},\mathbb{R}}(\lambda)$ is either $\mathbb{R}$ or the quaternion algebra $\mathbb{H}$, we see from Lemma 2.1(4) that D must be either $\mathbb{Q}$ or a quaternion algebra that does not split over $\mathbb{R}$.

Conversely, suppose D is either $\mathbb{Q}$ or a quaternion algebra that does not split over $\mathbb{R}$. In either case, D^2 is Brauer equivalent to $\mathbb{Q}$, so $D_{\mathfrak{g},\mathbb{Q}}(\lambda^i) \otimes_{\mathbb{Q}} \mathbb{R}$ is $\mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{R} = \mathbb{R}$ if i is even, and it is $D \otimes_{\mathbb{Q}} \mathbb{R} = \mathbb{H}$ if i is odd. This is a division algebra for every i, so we see from Lemma 2.1(3) that $\mathfrak{g}$ is $\mathbb{R}$-universal. □

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